

The axioms of n -exact categories

BACHELOR'S THESIS

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Introduction

Homological algebra is a key aspect of many modern topics in mathematics, be it geometry, algebra or topology. While originally only studied on modules and abelian groups, it was then later formalized on abelian categories, giving a more abstract framework. Exact categories, first introduced by D. Quillen, are a continuation of this abstraction, reducing the requirements to the bare minimum while still enabling talk about short exact sequences. In an attempt to also study longer exactness, a generalization to n -exact categories was thought of by G. Jasso. This paper aims to build the necessary tools to study such structures, while also providing a few examples.

First, we will generalize the structure of short exact sequences to additive categories, motivated by fundamental properties arising in abelian categories. We will then proceed with some examples of exact categories by finding extension-closed subcategories. Finally, we will give a smaller list of axioms that is equivalent to the classical definition.

Then, we will extend the concepts studied to longer exact sequences. We will define the necessary higher degree concepts in order to define n -exact categories, and proceed with a few elementary results. To conclude, we will give some examples of n -exact categories by looking at n -cluster tilting subcategories.

In order to understand this paper, the reader should have some knowledge about modern homological algebra. In particular, they should already be familiar with category theory, more specifically abelian category theory, as is presented for example in the lecture notes of MA3204 Homological Algebra [7].

For the last part of the paper, the reader should also be familiar with Auslander-Reiten theory and AR-quivers, as is for example explained in [1].

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Conventions and notations

Throughout this article, n always denotes a positive integer. $\mathcal{A}$ usually denotes an additive category. When not stated otherwise, the word kernel (respectively cokernel) will refer to the kernel morphism (respectively cokernel morphism) instead of the object. For a morphism $f : A \rightarrow B$, we call A its domain and B its codomain. We will use $\ker f$ to refer to its kernel morphism and $\operatorname{cok} f$ for its cokernel morphism, whereas $\operatorname{Ker} f$ will refer to its kernel object and $\operatorname{Cok} f$ its cokernel object. For any object $A \in \mathcal{A}$, we denote 1_A the identity morphism on A . We denote with $\operatorname{Ch}^n(\mathcal{A})$ the set of chain complexes on $\mathcal{A}$ concentrated in degrees $\{0, \dots, n\}$.

1 An introduction to exact categories

As a way to generalize the notion of abelian categories, one can look at a general additive category on which we define a certain notion of short exact sequences that has very similar properties to the abelian category counterpart. This chapter aims to motivate the definition of these so-called exact categories, to later define them formally and give a few concrete examples. We will also give a minimal version of the axioms due to B. Keller.

The content of this chapter is inspired by the lecture notes of MA3204 Homological Algebra [7], the article about exact categories by T. Bühler [3] and B. Keller's notes on exact categories [5, Appendix A].

1.1 Motivation

When studying abelian categories, there are many convenient properties related to monics and epics, which are used to prove results about short exact sequences. Let us recall a few:

Proposition 1.1 *In an abelian category,*

- *every monic is a kernel of its cokernel;*
- *every epic is a cokernel of its kernel.*

Furthermore, the following properties hold:

- *the pushout of a monic is a monic;*
- *the pullback of a monic is a monic;*
- *the pushout of an epic is an epic;*
- *the pullback of an epic is an epic.*

We refer the reader to [7, Sections 12-13] for the proofs of these facts.

Fundamentally, this tells us that every monic and every epic fits into a short exact sequence, and that taking the pullback or pushout of a morphism that is part of a short exact sequence gives rise to another short exact sequence. Indeed, if $f : A \rightarrowtail B$ is monic, then it fits into the short exact sequence $A \xrightarrow{f} B \xrightarrow{\text{cok}f} \text{Cok } f$. Furthermore, since morphisms that are part of a short exact sequence are either monic or epic, they will once again be monic or epic after a pullback or pushout, thus will once again fit in a short exact sequence.

Let us quickly recall what it means for a class of composable morphisms to be closed under isomorphisms.

Definition 1.2 Let $\mathcal{E}$ be a class of composable morphisms, i.e. a set of pairs (f, g) where the codomain of f is the domain of g . We say $\mathcal{E}$ is *closed under isomorphisms* if, for any commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ a \downarrow & & b \downarrow & & \downarrow c \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array}$$

where $(f, g) \in \mathcal{E}$ and the vertical maps are isomorphisms, $(f', g') \in \mathcal{E}$.

There are many properties we can conclude about kernels and cokernels using the previous results and this definition. These properties are what will motivate the definition of exact categories.

Proposition 1.3 *Let $\mathcal{A}$ be an abelian category and let $\mathcal{E}$ be the class of short exact sequences. $\mathcal{E}$ is closed under isomorphisms. Furthermore, the following hold:*

- [E0] *For any object $A \in \mathcal{A}$, the identity morphism 1_A is a kernel;*
- [E0^{op}] *For any object $A \in \mathcal{A}$, the identity morphism 1_A is a cokernel;*
- [E1] *The class of kernels is closed under composition;*
- [E1^{op}] *The class of cokernels is closed under composition;*

[E2] *The pushout of a kernel along any morphism is a kernel;*

[E2^{op}] *The pullback of a cokernel along any morphism is a cokernel.*

Proof It is clear that the class of short exact sequences is closed under isomorphisms. Indeed, $A \xrightarrow{f} B \xrightarrow{g} C$ is a short exact sequence if and only if f is a kernel of g and g is a cokernel of f . The result follows from the fact that kernels and cokernels are preserved by isomorphisms.

[E0] and [E0^{op}] follow directly from the fact that the identity morphism is both the kernel and cokernel of the zero map.

We can deduce [E1] and [E1^{op}] by recalling that composing monics (respectively epics) gives us monics and that monics are kernels of their cokernel (respectively epics are cokernels of their kernel).

As for [E2], recall that kernels are always monic. Thus, by 1.1, the pushout of a kernel is monic as well. But then, by 1.1 again, it is a kernel of its cokernel, thus it is a kernel. The proof for [E2^{op}] is similar. $\square$

Remark 1.5 We can in a very similar way prove that the pullback of a kernel (respectively the pushout of a cokernel) is a kernel (respectively cokernel), however we choose not to include this as it turns out to be unnecessary for many results.

1.2 Exact Categories

Based on the observations from the last section, one can define a more general notion that encapsulates the idea of short exact sequences, namely exact categories. To do so, let us first define kernel-cokernel pairs.

Definition 1.6 Let $\mathcal{A}$ be an additive category. A *kernel-cokernel pair* (i, p) in $\mathcal{A}$ is a pair of composable morphisms $A \xrightarrow{i} B \xrightarrow{p} C$ such that i is a kernel of p and p is a cokernel of i .

In the context of abelian categories, we notice that kernel-cokernel pairs indeed give rise to short exact sequences in the usual sense. In a general additive category, however, we cannot say the same since we have no guarantee that kernels and cokernels even exist for arbitrary morphisms, and thus we are unsure that $\text{Im } i$ exists. But we will use our intuition from abelian categories to expand the notion of short exact sequences to additive categories. First, let us fix some terminology.

Definition 1.7 Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$. We call a morphism $i : A \rightarrow B$ an *inflation* (or *admissible monic*) if there exists a morphism $p : B \rightarrow C$ such that $(i, p) \in \mathcal{E}$. Dually, we call a morphism $q : B \rightarrow C$ a *deflation* (or *admissible epic*) if there exists a morphism $j : A \rightarrow B$ such that $(j, q) \in \mathcal{E}$. We use $A \rightarrowtail B$ for inflations, and $B \twoheadrightarrow C$ for deflations. We say $\mathcal{E}$ is *closed under isomorphisms* if whenever we have $A \xrightarrow{i} B \xrightarrow{p} C \in \mathcal{E}$ fitting in a commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{p} & C \\ a \downarrow & & b \downarrow & & \downarrow c \\ A' & \xrightarrow{j} & B' & \xrightarrow{q} & C' \end{array}$$

with a , b , and c isomorphisms, then $(j, q) \in \mathcal{E}$.

It is usually convenient to require being closed under isomorphisms, as objects with universal properties are generally unique up to isomorphism. In particular, recall that all kernel objects of a specific morphism are isomorphic. In the case of a class of kernel-cokernel pairs $\mathcal{E}$, being closed under isomorphisms implies that all kernels of a deflation are inflations. Indeed, consider $B \twoheadrightarrow C$ a deflation and $A \rightarrowtail B$ an inflation such that $(i, p) \in \mathcal{E}$. Let $A' \xrightarrow{j} B$ be another kernel of p and $A \xrightarrow{a} A'$ be the isomorphism between both kernels. The sequence

$A' \xrightarrow{j} B \xrightarrow{p} C$ fits in the commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{p} & C \\ a \downarrow & & \parallel & & \parallel \\ A' & \xrightarrow{j} & B & \xrightarrow{p} & C \end{array}$$

Since all vertical arrows are isomorphisms, we conclude that $(j, p) \in \mathcal{E}$. The proof that all cokernels of an inflation are deflations is dual.

We can now proceed to define exact categories, basing our definition on the properties we established for abelian categories.

Definition 1.8 Let $\mathcal{A}$ be an additive category. An *exact structure* on $\mathcal{A}$ is a class $\mathcal{E}$ of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphisms and satisfies the following axioms:

- [E0] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an inflation;
- [E0^{op}] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is a deflation;
- [E1] The composition of two inflations yields an inflation;
- [E1^{op}] The composition of two deflations yields a deflation;
- [E2] The pushout of an inflation along any morphism yields an inflation;
- [E2^{op}] The pullback of a deflation along any morphism yields a deflation.

The last two axioms can be summarized by the following commutative diagrams:

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ c \downarrow & \lrcorner & \downarrow d \\ C' & \xrightarrow{f'} & D' \end{array} \quad \text{and} \quad \begin{array}{ccc} C' & \dashrightarrow & D' \\ \downarrow c & \lrcorner & \downarrow d \\ C & \xrightarrow{f} & D \end{array}$$

If $\mathcal{E}$ is an exact structure on $\mathcal{A}$, we say $(\mathcal{A}, \mathcal{E})$ is an *exact category* and the elements of $\mathcal{E}$ are called *conflations* (or *short exact sequences*).

1.3 Extension-closed subcategories

In 1.1, we saw that all abelian categories are exact categories, with a possible exact structure being all short exact sequences. We can generalize this example of exact categories to extension-closed subcategories of abelian categories. Let us first clarify what this means.

Definition 1.9 Let $\mathcal{A}$ be an abelian category and $\mathcal{C} \subseteq \mathcal{A}$ be a full additive subcategory of $\mathcal{A}$. $\mathcal{C}$ is said to be *closed under extension* if, for any short exact sequence in $\mathcal{A}$

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

such that A and C are both in $\mathcal{C}$, then B is also in $\mathcal{C}$.

In that context, we say that B is an *extension of C by A* .

Proposition 1.10 Let $\mathcal{A}$ be an abelian category and $\mathcal{C} \subseteq \mathcal{A}$ be a full additive subcategory of $\mathcal{A}$ that is closed under extensions. Let $\mathcal{E}$ be the collection of short exact sequences in $\mathcal{A}$ such that all three objects are in $\mathcal{C}$. Then $(\mathcal{C}, \mathcal{E})$ is an exact category.

Proof We prove each of the axioms one by one.

$\mathcal{E}$ is closed under isomorphisms since short exact sequences in $\mathcal{A}$ are closed under isomorphisms and $\mathcal{C}$ is full.

Since $\mathcal{C}$ is an additive subcategory, it contains the zero object. As such, for any object $B \in \mathcal{C}$, the sequences $0 \rightarrow B \xrightarrow{1_B} B$ and $B \xrightarrow{1_B} B \rightarrow 0$ are short exact and contained in $\mathcal{E}$, i.e. identity morphisms are both deflations and inflations.

Now let us consider the axiom [E2]. Let $X \xrightarrow{i} Y \xrightarrow{d} Z$ be a short exact sequence in $\mathcal{C}$ and $f : A \rightarrow X$. We can consider the pushout diagram

in the ambient category $\mathcal{A}$:

$$\begin{array}{ccccc} X & \xrightarrow{i} & Y & \xrightarrow{d} & Z \\ f \downarrow & & \downarrow f' & & \downarrow f'' \\ A & \xrightarrow{j} & B & \xrightarrow{e} & C \end{array}$$

where $C = \text{Cok } j$. From basic results in abelian categories, we know that j is monic, and that we can choose $C = Z$ (see [7, Proposition 13.7] for the latter). Thus the sequence $A \xrightarrow{j} B \xrightarrow{e} Z$ is short exact, with both A and Z in $\mathcal{C}$, which implies that $B \in \mathcal{C}$ since it is extension-closed. We therefore get a short exact sequence in $\mathcal{C}$, and j is an inflation. The proof for $[\text{E2}^{\text{op}}]$ is dual.

To prove $[\text{E1}]$, let us consider short exact sequences $A \xrightarrow{i} B \xrightarrow{d} C$ and $B \xrightarrow{j} Y \xrightarrow{e} Z$ in $\mathcal{C}$. Then we can construct the following pushout in the ambient category:

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \xrightarrow{j} & Y \\ & & \downarrow d & & \downarrow f \\ & & C & \xrightarrow{k} & E \end{array}$$

By 1.1, we know that f is epic. It then remains to prove that $j \circ i$ is a kernel of f . Since j is monic, one can show that the square is also a pullback (see [7, Corollary 13.6] for a proof). Consider $g : X \rightarrow Y$ such that $f \circ g = 0$. Using the pullback property on $g : X \rightarrow Y$ and $0 : X \rightarrow C$, we get a unique map $\hat{g} : X \rightarrow B$ such that $j \circ \hat{g} = g$ and $d \circ \hat{g} = 0$. Now using that i is a kernel of d , we get a unique map $\hat{\hat{g}} : X \rightarrow A$ such that $i \circ \hat{\hat{g}} = \hat{g}$. In particular, $(j \circ i) \circ \hat{\hat{g}} = g$, and $\hat{\hat{g}}$ is the unique map satisfying this property. Finally, we note that

$$f \circ (j \circ i) = (f \circ j) \circ i = (k \circ d) \circ i = k \circ (d \circ i) = 0$$

so that we can conclude that $j \circ i$ is a kernel of f . Since f is epic and $j \circ i$ is its kernel, we conclude that $A \xrightarrow{j \circ i} Y \xrightarrow{f} E$ is a short exact sequence

in $\mathcal{C}$. As such, inflations are closed under composition. Proving $[\mathbf{E1}^{\text{op}}]$ is dual. $\square$

Remark 1.12 A slightly modified converse statement is also true, and is known as the *Gabriel-Quillen embedding theorem*. It states that, if $(\mathcal{A}, \mathcal{E})$ is an exact category with $\mathcal{A}$ small, there is an embedding (i.e. a faithful functor that is injective on the objects) $\mathcal{A} \hookrightarrow \mathcal{B}$ into an abelian category such that $\mathcal{E}$ is the class of all short exact sequences in $\mathcal{B}$.

1.4 Examples

The previous statement is quite useful for finding exact categories that are not abelian categories. Here are a few fundamental general examples.

Examples 1.13

- Let R be a non-noetherian unital ring. Consider the category of finitely generated (left or right) R -modules $\text{mod}R$ (seen as a subcategory of $\text{Mod}R$). It is well known that this category is closed under extensions. However, it is not abelian. Indeed, consider a non-finitely generated ideal $I \subseteq R$ (which exists since R is non-noetherian). Then the kernel of the quotient map is $R \twoheadrightarrow R/I$ is the ideal I . In that case, both R and R/I are finitely generated R -modules, but I is by definition not finitely generated. Thus $\text{mod}R$ does not contain the kernel of all its maps, and cannot be abelian;
- Let R be any ring. Consider the full subcategory of $\text{mod}R$ consisting of all projectives $\mathcal{P}(R)$. Let $P, P' \in \mathcal{P}(R)$ be two projective R -modules. It is well known from homological algebra that $\text{Ext}_R^1(P', P) = 0$. But then, considering Ext_R^1 as the Yoneda extension functor, this implies that any short exact sequence $0 \rightarrow P \rightarrow E \rightarrow P' \rightarrow 0$ splits, i.e. $E = P \oplus P'$. Since the direct sum of two projectives is projective, this shows that $\mathcal{P}(R)$ is closed under extensions. However, it is not necessarily

abelian. For example, in the category $\text{mod}\mathbb{Z}$, the cokernel of the map $\mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}$ is $\mathbb{Z}/2\mathbb{Z}$, which is not projective, so $\mathcal{P}(\mathbb{Z})$ is not abelian.

1.5 Reducing the axioms

As was shown by B. Keller in [5, Appendix A], some of the axioms required in the classical definition of exact categories are redundant.

Proposition 1.14 *Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphisms. Assume that $\mathcal{E}$ satisfies $[\text{E1}^{\text{op}}]$, $[\text{E2}]$ as well as*

- $[\text{E0}']$ 1_0 is a deflation;
- $[\text{E2}^{\text{op}}']$ For any $f : C \rightarrow Z$ and deflation $d : Y \twoheadrightarrow Z$, there is a commutative square

$$\begin{array}{ccc} Y' & \xrightarrow{d'} \twoheadrightarrow & Z' \\ f' \downarrow & & \downarrow f \\ Y & \xrightarrow{d} \twoheadrightarrow & Z \end{array}$$

Then $\mathcal{E}$ is an exact structure on $\mathcal{A}$.

Proof We will first prove that $[\text{E2}]$ and $[\text{E2}^{\text{op}}']$, combined with closure under isomorphisms, imply $[\text{E2}^{\text{op}}]$ by explicitly building a pullback. Consider the commutative diagram from $[\text{E2}^{\text{op}}']$:

$$\begin{array}{ccccc} K & \xrightarrow{i'} & Y' & \xrightarrow{d'} \twoheadrightarrow & Z' \\ f_1 \downarrow & & f_2 \downarrow & & \downarrow f_3 \\ X & \xrightarrow{i} & Y & \xrightarrow{d} \twoheadrightarrow & Z \end{array}$$

f_1 exists as the map between kernels. As such we can build the pushout of i' along f_1 , which we will name P . This will be the pullback of d along

f_3 . To prove that, construct the maps $\hat{d}' : P \rightarrow Z'$ and $\hat{f}_2 : P \rightarrow Y$ by applying the pushout property to the following commutative diagrams:

$$\begin{array}{ccc}
 K & \xrightarrow{i'} & Y' \\
 f_1 \downarrow & \lrcorner & \downarrow \hat{f}_1 \\
 X & \xrightarrow{\hat{i}'} & P \\
 & \searrow \hat{d}' & \downarrow \\
 & & Z'
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 K & \xrightarrow{i'} & Y' \\
 f_1 \downarrow & \lrcorner & \downarrow \hat{f}_1 \\
 X & \xrightarrow{\hat{i}'} & P \\
 & \searrow \hat{f}_2 & \downarrow \\
 & & Y
 \end{array}$$

For P to be a pullback, it is necessary that $d \circ \hat{f}_2 = f_3 \circ \hat{d}'$ (since a pullback gives a commutative square). But this is true as both maps are a solution to the pushout problem

$$\begin{array}{ccccc}
 K & \xrightarrow{i'} & Y' & \xrightarrow{d'} & Z' \\
 f_1 \downarrow & & \downarrow \hat{f}_1 & & \downarrow f_3 \\
 X & \xrightarrow{\hat{i}'} & P & & \\
 \downarrow i & & \searrow \hat{f}_2 = f_3 \hat{d}' & & \downarrow \\
 Y & \xrightarrow{d} & Z & &
 \end{array}$$

Clearly $(d \circ i) \circ f_1 = 0 = f_3 \circ (d' \circ i')$, so that the big square commutes. Thus there exists a unique map $p : P \rightarrow Z$ such that $p \circ \hat{f}_1 = f_3 \circ d'$ and $p \circ \hat{i}' = d \circ i = 0$. But note that $f_3 \circ (\hat{d}' \circ \hat{f}_1) = f_3 \circ d'$ and $f_3 \circ (\hat{d}' \circ \hat{i}') = 0$ by definition of $\hat{d}'$, so $f_3 \circ \hat{d}' = p$. Similarly, $d \circ (\hat{f}_2 \circ \hat{f}_1) = d \circ f_2 = f_3 \circ d'$ by commutativity and $d \circ (\hat{f}_2 \circ \hat{i}') = d \circ i = 0$, so $d \circ \hat{f}_2 = p = f_3 \circ \hat{d}'$.

Next, we will show that $\hat{d}'$ is a cokernel of $\hat{i}'$. Consider a morphism $g : P \rightarrow D$ for some $D \in \mathcal{A}$ such that $g \circ \hat{i}' = 0$. Clearly, this implies that $g \circ (\hat{f}_1 \circ i') = g \circ (\hat{i}' \circ f_1) = 0$. Using that d' is a cokernel of i' , there exists a unique morphism $k : Z' \rightarrow D$ such that $k \circ d' = g \circ \hat{f}_1$. What is left to show is that $k \circ \hat{d}' = g$, but this is true since they are

both solutions to the pushout problem

$$\begin{array}{ccccc}
 K & \xrightarrow{i'} & Y' & & \\
 f_1 \downarrow & & \hat{f}_1 \downarrow & \searrow & \\
 X & \xrightarrow{\hat{i}'} & P & \xrightarrow{\hat{f}_1 g = k d'} & D \\
 & \searrow \hat{i}' & & \nwarrow k \hat{d}' = g & \\
 & & 0 & \searrow & \\
 & & & & D
 \end{array}$$

Therefore, there exists a unique map k such that $k \circ \hat{d}' = g$, i.e. $\hat{d}'$ is a cokernel of $\hat{i}'$. This implies the top row in the commutative diagram

$$\begin{array}{ccccc}
 X & \xrightarrow{\hat{i}'} & P & \xrightarrow{\hat{d}'} & Z' \\
 \parallel & & \hat{f}_2 \downarrow & & f_3 \downarrow \\
 X & \xrightarrow{i} & Y & \xrightarrow{d} & Z
 \end{array}$$

is a conflation, which is a requirement in [E2^{op}].

The last step is to prove that the rightmost square is a pullback. To do this, we will need to create a few more morphisms. Consider the pushout of $\hat{i}'$ along i

$$\begin{array}{ccccc}
 X & \xrightarrow{\hat{i}'} & P & \xrightarrow{\hat{d}'} & Z' \\
 i \downarrow & \swarrow \hat{f}_2 & \downarrow j & & \parallel \\
 Y & \xrightarrow{\hat{j}'} & C & \xrightarrow{\hat{e}'} & Z' \\
 \downarrow d & & \downarrow e & & \\
 Z & = & Z & &
 \end{array}$$

where all rows and columns are conflations due to [E2] (it is not hard to show that one can choose Z' (respectively Z) as the cokernel object of $\hat{j}'$ (respectively j) using the pushout property). Considering the

pushout property applied to the commutative diagram

$$\begin{array}{ccc}
 X & \xrightarrow{\hat{i}'} & P \\
 \downarrow i & & \downarrow j \\
 Y & \xrightarrow{\hat{j}'} & C
 \end{array}
 \quad
 \begin{array}{c}
 \searrow \hat{f}_2 \\
 \quad \quad \quad \searrow \exists! r \\
 \quad \quad \quad Y
 \end{array}$$

we get a unique map $r : C \rightarrow Y$ such that $r \circ \hat{j}' = 1_Y$ and $r \circ j = \hat{f}_2$. Since

$$(1_C - \hat{j}' \circ r) \circ \hat{j}' = \hat{j}' - \hat{j}' \circ (r \circ \hat{j}') = \hat{j}' - \hat{j}' = 0,$$

we get a unique map $g : Z' \rightarrow C$ such that $g \circ \hat{d}' = 1_C - \hat{j}' \circ r$.

Note that $-e \circ g = f_3$, since both $f_3 \circ \hat{e}'$ and $-e \circ g \circ \hat{e}'$ are solutions of the pushout problem

$$\begin{array}{ccc}
 X & \xrightarrow{\hat{i}'} & P \\
 \downarrow i & & \downarrow j \\
 Y & \xrightarrow{\hat{j}'} & C
 \end{array}
 \quad
 \begin{array}{c}
 \searrow f_3 \hat{d}' \\
 \quad \quad \quad \searrow -e g \hat{e}' = f_3 \hat{e}' \\
 \quad \quad \quad Z
 \end{array}$$

Indeed, that both compositions with $\hat{j}'$ are zero is direct since $(\hat{j}', \hat{e}')$ is a conflation. And since $-e \circ (g \circ \hat{e}') = -e \circ (1_C - \hat{j}' \circ r) = -e + d \circ r$, we get that

$$(-e \circ g \circ \hat{e}') \circ j = (-e + d \circ r) \circ j = -e \circ j + d \circ b = f_3 \circ \hat{d}'.$$

so that it solves the pushout problem. Similarly, $f_3 \circ (\hat{e}' \circ j) = f_3 \circ \hat{d}'$, so it solves the pushout problem as well. Thus, $-e \circ g \circ \hat{e}' = f_3 \circ \hat{e}'$ and therefore $-e \circ g = f_3$ since $\hat{e}'$ is epic.

Now we are ready to show that P is the pullback of d along f_3 . Consider the commutative diagram

$$\begin{array}{ccc}
 D & \xrightarrow{z} & Z' \\
 \searrow y & & \downarrow f_3 \\
 P & \xrightarrow{\hat{d}'} & Z' \\
 \downarrow \hat{f}_2 & & \downarrow f_3 \\
 Y & \xrightarrow{d} & Z
 \end{array}$$

Note that $f_3 \circ z = d \circ y$ if and only if $e \circ (\hat{j}' \circ y + g \circ z) = 0$ (using that $d = e \circ \hat{j}'$ and $f_3 = -e \circ g$). Thus, using that j is a kernel of e , we get a unique map $h : D \rightarrow P$ such that $j \circ h = \hat{j}' \circ y + g \circ z$. Then, note that $r \circ g \circ \hat{e}' = r \circ (1_C - \hat{j}' \circ r) = r - r = 0$, and therefore $r \circ g = 0$. Thus,

$$\hat{f}_2 \circ h = r \circ j \circ h = r \circ \hat{j}' \circ y + r \circ g \circ z = y.$$

In addition, since $\hat{e}' \circ g \circ \hat{e}' = \hat{e}' \circ (1_C - \hat{j}' \circ r) = \hat{e}'$ as $(\hat{j}', \hat{e}')$ is a conflation, we get that $\hat{e}' \circ g = 1_{Z'}$. Finally,

$$\hat{d}' \circ h = \hat{e}' \circ j \circ h = \hat{e}' \circ \hat{j}' \circ y + \hat{e}' \circ g \circ z = z.$$

Thus, we get that $\hat{d}' \circ h = z$ and $\hat{f}_2 \circ h = y$, i.e. h solves the pullback problem. This allows us to conclude that the square

$$\begin{array}{ccc}
 P & \xrightarrow{\hat{d}'} & Z' \\
 \hat{f}_2 \downarrow & & \downarrow f_3 \\
 Y & \xrightarrow{d} & Z
 \end{array}$$

is indeed a pullback, and shows that $[\text{E2}^{\text{op}}]$ holds.

We will not prove that, if $[\text{E0}']$, $[\text{E1}^{\text{op}}]$, $[\text{E2}]$ and $[\text{E2}^{\text{op}}]$ hold, then so does $[\text{E1}]$, but refer the reader to [5, Appendix A.1].

Thus it only remains to show that $[E0']$ implies both $[E0]$ and $[E0^{op}]$. First, note that 1_0 is also an inflation, since it is its own kernel, and it is a deflation. Next, note that the commutative square

$$\begin{array}{ccc} X & \xrightarrow{1_X} & X \\ 0 \downarrow & \lrcorner & \downarrow 0 \\ 0 & \xrightarrow{0} & 0 \end{array}$$

is a pullback. Indeed, let $D \in \mathcal{A}$ fit into the commutative diagram

$$\begin{array}{ccccc} D & & & & \\ & \searrow g & & & \\ & & X & \xrightarrow{1_X} & X \\ & \searrow 0 & \downarrow 0 & & \downarrow 0 \\ & & 0 & \xrightarrow{0} & 0 \end{array}$$

Then the only morphism that can fit as the dashed arrow is g itself, and the pullback property is trivially fulfilled. Therefore, because of $[E2^{op}]$, 1_X is a deflation. Similarly, but using a pushout instead, 1_X is also an inflation. This proves that both $[E0]$ and $[E0^{op}]$ are true. $\square$

2 Expanding to n -exact categories

After generalizing from abelian categories to exact categories, we can go one step further by looking at longer exact sequences. First, we will redefine the notions needed for exact categories, but in a more general context. Then we will proceed to define n -exact categories, and talk about a few properties. Finally, we will give some concrete examples of such categories.

The contents of this chapter are based on [4].

2.1 Working in higher degrees

Before defining n -exact categories, there are many prerequisites that need to be addressed. Firstly, we will need a way to properly describe what the equivalent of a kernel-cokernel pair would be in this new context. As a result, it makes sense to define n -kernels and n -cokernels, which require the notion of weak kernel (respectively weak cokernel).

Definition 2.1 Let $f : A \rightarrow B$ be a morphism in $\mathcal{A}$. A *weak kernel* of f is an object $K \in \mathcal{A}$ with a morphism $k : K \rightarrow A$ such that the sequence of abelian groups

$$\mathcal{A}(C', K) \xrightarrow{k \circ -} \mathcal{A}(C', A) \xrightarrow{f \circ -} \mathcal{A}(C', B)$$

is exact for all $C' \in \mathcal{A}$.

Dually, a *weak cokernel* of f is an object $C \in \mathcal{A}$ with a morphism $c : B \rightarrow C$ such that the sequence of abelian groups

$$\mathcal{A}(C, C') \xrightarrow{- \circ c} \mathcal{A}(B, C') \xrightarrow{- \circ f} \mathcal{A}(A, C')$$

is exact for all $C' \in \mathcal{A}$.

Remarks 2.2 We do not ask for the sequence to be short exact, but only for it to be exact in the middle term. Furthermore, the following observations are quite useful in practice:

- (a) Notice the direction of the sequences is different since for weak cokernels, we apply the contravariant functor $\mathcal{A}(-, C')$ to our original sequence, whereas we apply the covariant functor $\mathcal{A}(C', -)$ in the case of weak kernels;
- (b) For $k : K \rightarrow A$ a weak kernel of $f : A \rightarrow B$, the sequence being exact lets us know that
 - (i) The composition $f \circ k$ is 0;
 - (ii) For any morphism $h : C \rightarrow A$ such that $f \circ h = 0$, there exists a morphism $\hat{h} : C \rightarrow K$ such that $k \circ \hat{h} = h$.

That last point can be summarized by the following diagram:

$$\begin{array}{ccccc}
 K & \xrightarrow{k} & A & \xrightarrow{f} & B \\
 \uparrow \exists \hat{h} & & \nearrow h & \searrow 0 & \\
 C & & & &
 \end{array}$$

These two points follow from exactness in the middle term. Indeed, since $\text{Im} (k \circ -) = \text{Ker} (f \circ -)$, any map $h : C \rightarrow A$ such that $f \circ h = 0$, is in the image of the post-composition with k , i.e. can be factored (not necessarily uniquely!) as $k \circ \hat{h}$. In particular, if we consider $C = K$, we conclude that

$$k = k \circ 1_K \in \text{Im} (k \circ -) = \text{Ker} (f \circ -),$$

i.e. $f \circ k = 0$. The result for weak cokernels is dual.

In summary, weak cokernels and weak kernels have a very similar property to that of cokernels and kernels, except that uniqueness in our

factoring of morphisms is not guaranteed. In other words, the only difference is that a weak kernel is not necessarily monic, and a weak cokernel is not necessarily epic.

This notion is quite fundamental in the elaboration of n -kernels and n -cokernels.

Definition 2.3 Let $d^0 : X^0 \rightarrow X^1$ be a morphism in $\mathcal{A}$. An n -kernel of d^0 is a sequence

$$X^{-n} \xrightarrow{d^{-n}} \dots \xrightarrow{d^{-1}} X^0 \left(\xrightarrow{d^0} X^1 \right)$$

such that the sequence induced by applying the functor $\mathcal{A}(C', -)$

$$0 \rightarrow \mathcal{A}(C', X^{-n}) \xrightarrow{d^{-n} \circ -} \mathcal{A}(C', X^{-n+1}) \xrightarrow{d^{-n+1} \circ -} \dots \xrightarrow{d^0 \circ -} \mathcal{A}(C', X^1)$$

is exact.

Equivalently, an n -kernel of d^0 is such a sequence $(d^{-n}, d^{-n+1}, \dots, d^{-1})$, where d^{-k} is a weak cokernel of d^{-k+1} for all $1 \leq k \leq n-1$, and d^{-n} is a kernel of d^{-n+1} .

Dually, an n -cokernel of d^0 is a sequence

$$X^1 \xrightarrow{d^1} X^2 \xrightarrow{d^2} \dots \xrightarrow{d^n} X^{n+1}$$

such that d^k is a weak cokernel of d^{k-1} for $1 \leq k \leq n-1$ and d^n is a cokernel of d^{n-1} .

Remark 2.4 In the case $n = 1$, this indeed coincides with the usual definition of kernels and cokernels.

Now that we have a suitable notion of higher kernels and cokernels, we can consider the concept of n -exact sequences in the most logical way.

Definition 2.5 An n -exact sequence in $\mathcal{A}$ is a chain complex

$$X^0 \xrightarrow{d^0} X^1 \xrightarrow{d^1} \dots \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1}$$

such that $(d^0, \dots, d^{n-1})$ is an n -kernel of d^n and $(d^1, \dots, d^n)$ is an n -cokernel of d^0 .

Remark 2.6 Once again, when $n = 1$, we get back to our usual notion of short exact sequence.

There are two final notions that need to be covered before proceeding with the definition of n -exact categories: that of n -pullback, n -pushout, and weak isomorphisms. Before covering these concepts, let us first recall what the cone of a morphism of complexes is:

Definition 2.7 Let X and Y be two complexes in $\text{Ch}(\mathcal{A})$ and $f : X \rightarrow Y$ be a morphism of complexes, or diagrammatically:

$$\begin{array}{ccccccc} X : & \dots & \longrightarrow & X^{k-1} & \xrightarrow{d_X^{k-1}} & X^k & \xrightarrow{d_X^k} & X^{k+1} & \longrightarrow & \dots \\ & & & \downarrow f^{k-1} & & \downarrow f^k & & \downarrow f^{k+1} & & \\ Y : & \dots & \longrightarrow & Y^{k-1} & \xrightarrow{d_Y^{k-1}} & Y^k & \xrightarrow{d_Y^k} & Y^{k+1} & \longrightarrow & \dots \end{array}$$

The *cone of f* , denoted $C(f)$, is the complex

$$C(f) : \quad \dots \xrightarrow{d_C^{k-2}} X^k \oplus Y^{k-1} \xrightarrow{d_C^{k-1}} X^{k+1} \oplus Y^k \xrightarrow{d_C^k} \dots$$

where $d_C^k = \begin{bmatrix} -d_X^{k+1} & 0 \\ f^{k+1} & d_Y^k \end{bmatrix}$.

Before looking at the higher equivalents of pullbacks and pushouts, it is quite useful to examine the link that they have with the cone of a map.

Remark 2.8 Let $B^1 \xrightarrow{d_B^1} B^2$ and $A^2 \xrightarrow{f^2} B^2$ be two morphisms in $\mathcal{A}$ and consider the pullback of d_B^1 along f^2 :

$$\begin{array}{ccc} A^1 & \xrightarrow{d_A^1} & A^2 \\ f^1 \downarrow & \lrcorner & \downarrow f^2 \\ B^1 & \xrightarrow{d_B^1} & B^2 \end{array}$$

Clearly $f = (f^1, f^2)$ is a morphism of complexes since the square commutes. Thus we can consider its cone:

$$C(f) : \quad 0 \longrightarrow A^1 \xrightarrow{d_C^0} A^2 \oplus B^1 \xrightarrow{d_C^1} B^2 \longrightarrow 0$$

where $d_C^0 = \begin{bmatrix} -d_A^1 \\ f^1 \end{bmatrix}$, and $d_C^1 = \begin{bmatrix} f^2 & d_B^1 \end{bmatrix}$. We claim that d_C^0 is a kernel of d_C^1 . Indeed, consider a morphism $\begin{bmatrix} g \\ h \end{bmatrix} : C \rightarrow A^2 \oplus B^1$ such that the composition $d_C^1 \circ \begin{bmatrix} g \\ h \end{bmatrix} = 0$. The composition being 0 implies that $f^2 \circ g + d_B^1 \circ h = 0$, or $d_B^1 \circ h = -f^2 \circ g$. Consider the unique map $\hat{h} : C \rightarrow A^1$ given by $f^1 \circ \hat{h} = h$ and $d_A^1 \circ \hat{h} = -g$ (this map comes from the pullback property). We see that $d_C^0 \circ \hat{h} = \begin{bmatrix} -d_A^1 \circ \hat{h} \\ f^1 \circ \hat{h} \end{bmatrix} = \begin{bmatrix} g \\ h \end{bmatrix}$. Thus we have defined a unique map $\hat{h} : C \rightarrow A^1$ such that $d_C^0 \circ \hat{h} = \begin{bmatrix} g \\ h \end{bmatrix}$. That is, d_C^0 is a kernel of d_C^1 .

A dual result is true for pushouts, where d_C^1 is a cokernel of d_C^0 .

Basing ourselves on this observation, we can generalize the notion of pullback and pushout to higher degrees:

Definition 2.9 Let $X : X^0 \xrightarrow{d_X^0} X^1 \xrightarrow{d_X^1} \dots \xrightarrow{d_X^{n-1}} X^n$ be a complex in $\text{Ch}^n(\mathcal{A})$ and $f^0 : X^0 \rightarrow Y^0$ a morphism in $\mathcal{A}$. An n -pushout of X along f^0 is a morphism of complexes

$$\begin{array}{ccccccc} X : & X^0 & \xrightarrow{d_X^0} & X^1 & \xrightarrow{d_X^1} & \dots & \xrightarrow{d_X^{n-1}} X^n \\ & \downarrow f & & \downarrow f^0 & & \downarrow f^1 & & \downarrow f^n \\ Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \dots & \xrightarrow{d_Y^{n-1}} Y^n \end{array}$$

such that in the cone of f

$$C(f) : \quad X^0 \xrightarrow{d_C^{-1}} X^1 \oplus Y^0 \xrightarrow{d_C^0} \dots \xrightarrow{d_C^{n-1}} X^n \oplus Y^{n-1} \xrightarrow{d_C^n} Y^n$$

$(d_C^0, \dots, d_C^n)$ is an n -cokernel of d_C^{-1} .

The definition for the n -pullback of a chain complex is dual. That is, given a complex Y in $\text{Ch}^n(\mathcal{A})$ and a morphism $f^n : X^n \rightarrow Y^n$, an n -pullback of Y along f^n is a morphism of complexes $f : X \rightarrow Y$ such that in its cone $C = C(f)$, the sequence $(d_C^{-1}, \dots, d_C^{n-1})$ is an n -kernel of d_C^n .

The last notion required for defining n -exact categories is that of weak isomorphisms:

Definition 2.10 Let X and Y be two n -exact sequences in $\mathcal{A}$. We say that $f : X \rightarrow Y$ is a *weak isomorphism* of complexes if there is $k \in \{0, 1, \dots, n\}$ such that f^k and f^{k+1} are isomorphisms.

When $n = 1$, we note that weak isomorphisms are isomorphisms of short exact sequences. This can be shown by using the five lemma, which holds for any exact category (see [3, Lemma 8.9]). In the general case, however, weak isomorphisms are not necessarily isomorphisms, but a weaker result still holds.

Proposition 2.11 Let X and Y be two n -exact sequences in $\mathcal{A}$ and let $f : X \rightarrow Y$ be a morphism of complexes. If f^k and f^{k+1} are iso-

morphisms for some $k \in \{0, 1, \dots, n\}$, then f induces an isomorphism in $H(\mathcal{A})$.

A proof of this result can be found in [4, Proposition 2.7].

All of this preliminary work being done, we can finally proceed to define n -exact categories.

2.2 n -exact categories

With the proper tools put in place, it is quite straightforward to expand the definition of exact categories to longer exact sequences. Similar to what was done in chapter 1, we proceed with admissible n -exact sequences, admissible monics, and admissible epics.

Definition 2.12 Let $\mathcal{E}$ be a class of n -exact sequences in $\mathcal{A}$. An $\mathcal{E}$ -admissible n -exact sequence is an n -exact sequence that is in $\mathcal{E}$. An $\mathcal{E}$ -admissible monic is a morphism $d_X^0 : X^0 \rightarrow X^1$ such that there is an $\mathcal{E}$ -admissible n -exact sequence

$$X^0 \xrightarrow{d_X^0} X^1 \xrightarrow{d_X^1} \dots \xrightarrow{d_X^n} X^{n+1}.$$

Dually, an $\mathcal{E}$ -admissible epic is a morphism $d_X^n : X^n \rightarrow X^{n+1}$ such that there is an $\mathcal{E}$ -admissible n -exact sequence

$$X^0 \xrightarrow{d_X^0} \dots \xrightarrow{d_X^{n-1}} X^n \xrightarrow{d_X^n} X^{n+1}.$$

Whenever the context is clear, we will simply say admissible n -exact sequence (respectively admissible monic, admissible epic). We shall write $X^0 \xrightarrow{d_X^0} X^1$ for admissible monics and $X^n \xrightarrow{d_X^n} X^{n+1}$ for admissible epics.

The following definition simply adapts the axioms of exact categories to the definitions given in the previous section:

Definition 2.13 Let $\mathcal{E}$ be a class of n -exact sequences on $\mathcal{A}$. $\mathcal{E}$ is said to be an n -exact structure on $\mathcal{A}$ if it is closed under weak isomorphisms and it satisfies the following axioms:

- [E0] The sequence $0 \rightharpoonup 0 \rightarrow \cdots \rightarrow 0$ is an $\mathcal{E}$ -admissible n -exact sequence;
- [E1] The composition of two $\mathcal{E}$ -admissible monics yields an $\mathcal{E}$ -admissible monic;
- [E1^{op}] The composition of two $\mathcal{E}$ -admissible epics yields an $\mathcal{E}$ -admissible epic;
- [E2] If $X : X^0 \xrightarrow{d_X^0} \cdots \xrightarrow{d_X^n} X^{n+1}$ is an $\mathcal{E}$ -admissible n -exact sequence and $X^0 \xrightarrow{f^0} Y^0$ is a morphism in $\mathcal{A}$, there exists an n -pushout of $(d_X^0, \dots, d_X^{n-1})$ along f^0

$$\begin{array}{ccccccc}
 X : & X^0 & \xrightarrow{d_X^0} & X^1 & \xrightarrow{d_X^1} & \cdots & \xrightarrow{d_X^{n-1}} X^n \cdots \cdots \gg X^{n+1} \\
 \downarrow f & \downarrow f^0 & & \downarrow f^1 & & & \downarrow f^n \\
 Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \cdots & \xrightarrow{d_Y^{n-1}} Y^n
 \end{array}$$

such that d_Y^0 is an $\mathcal{E}$ -admissible monic;

- [E2^{op}] If $Y : Y^0 \xrightarrow{d_Y^0} \cdots \xrightarrow{d_Y^n} Y^{n+1}$ is an $\mathcal{E}$ -admissible n -exact sequence and $X^{n+1} \xrightarrow{f^{n+1}} Y^{n+1}$ is a morphism in $\mathcal{A}$, there is an n -pullback of $(d_Y^1, \dots, d_Y^n)$ along f^{n+1}

$$\begin{array}{ccccccc}
 X : & & & X^1 & \xrightarrow{d_X^1} & \cdots & \xrightarrow{d_X^{n-1}} X^n \xrightarrow{d_X^n} \gg X^{n+1} \\
 \downarrow f & & & \downarrow f^1 & & & \downarrow f^{n+1} \\
 Y : & Y^0 & \xrightarrow{d_Y^0} & Y^1 & \xrightarrow{d_Y^1} & \cdots & \xrightarrow{d_Y^{n-1}} Y^n \xrightarrow{d_Y^n} \gg Y^{n+1}
 \end{array}$$

such that d_X^n is an $\mathcal{E}$ -admissible epic.

If $\mathcal{E}$ is an n -exact structure on $\mathcal{A}$, we say that $(\mathcal{A}, \mathcal{E})$ is an n -exact category.

Remarks 2.14 • The axiom [E0] given here seems to be less restrictive than the [E0] and [E0^{op}] of exact categories. However, if $n \geq 2$, note that identity morphisms are also admissi-

ble monics since

$$\begin{array}{ccccccc}
 X & \xrightarrow{1_X} & X & \longrightarrow & 0 & \longrightarrow & \cdots \longrightarrow 0 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \longrightarrow 0
 \end{array}$$

is a weak isomorphism. Identity morphisms are admissible epics for the same reason.

- One can show that n -kernels and n -cokernels are unique up to weak isomorphism. As such, all n -kernels (respectively n -cokernels) of an admissible epic (respectively admissible monic) fit into an admissible n -exact sequence.

Most classical results from exact categories can be generalized to n -exact categories without much trouble. For example, there is a way to generalize Quillen's obscure axiom.

Proposition 2.15 (Obscure axiom) *Let $(\mathcal{A}, \mathcal{E})$ be an n -exact category. Suppose there is a commutative diagram*

$$\begin{array}{ccccccc}
 X : & X^0 & \xrightarrow{d_X^0} & \cdots & \xrightarrow{d_X^{n-1}} & X^n & \xrightarrow{d_X^n} \twoheadrightarrow X^{n+1} \\
 \downarrow f & \downarrow f^0 & & & \downarrow f^n & & \parallel \\
 Y : & Y^0 & \xrightarrow{d_Y^0} & \cdots & \xrightarrow{d_Y^{n-1}} & Y^n & \xrightarrow{d_Y^n} Y^{n+1}
 \end{array}$$

where the top row is an $\mathcal{E}$ -admissible n -exact sequence and $(d_Y^0, \dots, d_Y^{n-1})$ is an n -kernel of d_Y^n . Then Y is an $\mathcal{E}$ -admissible n -exact sequence.

A proof of the dual statement can be found in [4, Proposition 4.11]

Remark 2.16 What is really required for the statement to hold is a morphism $f^n : X^n \rightarrow Y^n$ such that $d_X^n = d_Y^n f^n$. Indeed, such a map can be extended to a morphism between n -kernels in much the

same way that it induces a morphism between kernels in the exact case. Since $d_Y^n(f^n d_X^{n-1}) = (d_Y^n f^n) d_X^{n-1} = d_X^n d_X^{n-1} = 0$, there exists a (not necessarily unique) morphism $f^{n-1} : X^{n-1} \rightarrow Y^{n-1}$ such that $d_Y^{n-1} f^{n-1} = f^n d_X^{n-1}$, since d_Y^{n-1} is a weak kernel of d_Y^n . Playing the same game again with f^{n-1} , we can create a morphism of complexes $f : X \rightarrow Y$, giving us the required setup.

A convenient property of n -exact categories is that admissible n -exact sequences interact nicely with direct sums. This gives an easy way to construct new n -exact sequences.

Proposition 2.17 *Let $(\mathcal{A}, \mathcal{E})$ be an n -exact category and $X, Y \in \text{Ch}^{n+1}(\mathcal{A})$. Then X and Y are admissible n -exact sequences if and only if $X \oplus Y$ is an admissible n -exact sequence.*

Proof The proof is inspired by those of [4, Proposition 4.6] and [4, Proposition 4.12].

Assume both X and Y are admissible n -exact sequences. Consider the sequence

$$X^0 \oplus Y^0 \xrightarrow{d_X^0 \oplus 1_{Y^0}} X^1 \oplus Y^0 \xrightarrow{\begin{bmatrix} d_X^1 & 0 \end{bmatrix}} X^2 \xrightarrow{d_X^2} \dots \xrightarrow{d_X^{n-1}} X^n \xrightarrow{d_X^n} X^{n+1}.$$

Since it is different of X only by a contractible complex, it is weakly isomorphic to it, and therefore an admissible n -exact sequences. Thus $d_X^0 \oplus 1_{Y^0}$ is an admissible monic. Similarly, $1_{X^1} \oplus d_Y^0$ is an admissible monic. Thus, because of [E1],

$$d_X^0 \oplus d_Y^0 = (1_{X^1} \oplus d_Y^0) \circ (d_X^0 \oplus 1_{Y^0})$$

is an admissible monic, and $(d_X^1 \oplus d_Y^1, \dots, d_X^n \oplus d_Y^n)$ is one of its n -cokernels. As such, $X \oplus Y$ is a admissible n -exact sequence.

Conversely, assume $X \oplus Y$ is an admissible n -exact sequence. Then, we can exhibit $[d_X^n \ 0] = [1_{X^{n+1}} \ 0] \circ (d_X^n \oplus d_Y^n)$ (which is an admissible epic by [E1^{op}]) as

$$\begin{array}{ccccc}
 X^n \oplus Y^n & \xrightarrow{d_X^n \oplus d_Y^n} & X^{n+1} \oplus Y^{n+1} & \xrightarrow{[1_{X^{n+1}} \ 0]} & X^{n+1} \\
 \downarrow [1_X^n \ 0] & & & & \parallel \\
 X^n & \xrightarrow{d_X^n} & & & X^{n+1}
 \end{array}$$

Thus d_X^n is an admissible epic by 2.15, and admits as n -kernel $(d_X^0, \dots, d_X^{n-1})$, i.e. X is an admissible n -exact sequence. The same can be said about Y . $\square$

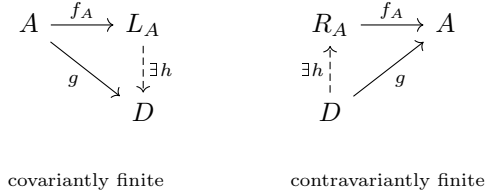
2.3 n -cluster tilting subcategories

Similar to extension-closed subcategories of abelian categories, there is a higher degree notion that gives us natural examples of n -exact categories. But this requires a few more definitions, namely what being functorially finite means.

Definition 2.19 Let $\mathcal{A}$ be an additive category and $\mathcal{D}$ be a full subcategory of $\mathcal{A}$. $\mathcal{D}$ is called *covariantly finite* if for any object $A \in \mathcal{A}$ there exists an object $L_A \in \mathcal{D}$ and a morphism $f_A : A \rightarrow L_A$ such that for any morphism $g : A \rightarrow D$ where $D \in \mathcal{D}$, there exists a morphism $h : L_A \rightarrow D$ such that $g = h \circ f_A$.

Dually, $\mathcal{D}$ is called *contravariantly finite* if for any object $A \in \mathcal{A}$ there exists an object $R_A \in \mathcal{D}$ and a morphism $f_A : R_A \rightarrow A$ such that for any morphism $g : D \rightarrow A$ where $D \in \mathcal{D}$, there exists a morphism $h : D \rightarrow R_A$ such that $g = f_A \circ h$.

Both properties can be summarized in the following diagrams:



$\mathcal{D}$ is then said to be *functorially finite* if it is both covariantly and contravariantly finite.

Remark 2.20 In the context of module categories, this is very often true for any subcategory. In fact, if A is an Artin algebra, one can show that every subcategory of $\text{mod } A$ is functorially finite if and only if A is of finite representation type (see [2, Proposition 1.2]).

By adding some criteria on how the Ext functor acts with the subcategory, we come to the definition of n -cluster tilting subcategories.

Definition 2.21 Let $\mathcal{A}$ be an abelian category and $\mathcal{D}$ be a full subcategory of $\mathcal{A}$. $\mathcal{D}$ is an n -cluster tilting subcategory of $\mathcal{A}$ if

- it is functorially finite;
- $\mathcal{D} = \{C \in \mathcal{A} \mid \text{Ext}_{\mathcal{A}}^i(C, D) = 0 \ \forall D \in \mathcal{D} \ \forall i \in \{1, \dots, n-1\}\}$;
- $\mathcal{D} = \{C \in \mathcal{A} \mid \text{Ext}_{\mathcal{A}}^i(D, C) = 0 \ \forall D \in \mathcal{D} \ \forall i \in \{1, \dots, n-1\}\}$.

Finding n -cluster tilting categories is a good way to create examples of n -exact categories, as was proven in [4, Theorem 3.16]:

Theorem 2.22 Let $\mathcal{A}$ be an abelian category and $\mathcal{D}$ be an n -cluster tilting subcategory of $\mathcal{A}$. Then $\mathcal{D}$ is an n -exact category with the n -exact structure given by the class of all n -exact sequences in $\mathcal{D}$.

Remarks 2.23 • G. Jasso actually proved a stronger result, stating that $\mathcal{D}$ is even an n -abelian category. However, we will not go further into this, and only refer to [4, Section 3.1];

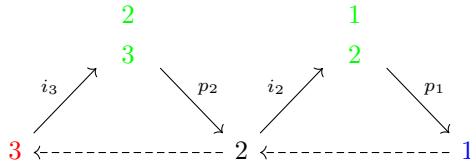
- Under a few extra assumptions, the converse statement also holds (see COULDN'T FIND THE SOURCE);
- A similar theory can be built on top of exact categories instead of abelian categories, which leads to obtaining n -exact categories that are not necessarily n -abelian (see [4, Section 4.2]). Again, a converse statement of sorts also exists, see [6, Theorem 5.3].

2.4 Building concrete examples from AR-quivers of path algebras

When working in sufficiently nice module categories, it is quite easy to prove a subcategory is n -cluster tilting by working on the AR-quiver and interpreting the Ext_Λ^i -functor as i -extensions.

Example 2.24 For the sake of simplicity, there are technical details left unaddressed.

Consider $\Lambda = k(1 \xrightarrow{\alpha} 2 \xrightarrow{\beta} 3) / \langle \beta\alpha \rangle$. Its AR-quiver is simply



The green modules are indecomposable projective-injective (from left to right $P_2 = I_2$ and $P_1 = I_3$), the red one is an indecomposable projective (P_3) and the blue one is an indecomposable injective (I_1).

Consider the full subcategory $\mathcal{M}$ of $\text{mod } \Lambda$ given by the additive closure of the colored indecomposables, i.e. $M \in \mathcal{M}$ if and only if $M = P_3^{r_1} \oplus P_2^{r_2} \oplus P_1^{r_3} \oplus I_1^{r_4}$.

First, it is functorially finite since Λ is an Artin algebra of finite representation type. Let us show it is 2-cluster tilting. To do so, one must show that both $\text{Ext}_\Lambda^1(M, \mathcal{M}) = 0$ and $\text{Ext}_\Lambda^1(\mathcal{M}, M) = 0$ for any $M \in \mathcal{M}$, and that both are non-zero if $M \notin \mathcal{M}$. But this can be deduced graphically for the indecomposables: the only possible 3-element sequences in $\mathcal{M}$ with non-zero maps are $P_3 \xrightarrow{i_3} P_2 \xrightarrow{i_2 p_2} P_1$ and $P_2 \xrightarrow{i_2 p_2} P_1 \xrightarrow{p_2} I_1$. However, these are certainly not exact, if only for the fact that $l(P_3) + l(P_1) = 3 \neq 2 = l(P_2)$ and $l(P_2) + l(P_1) = 3 \neq 2 = l(P_1)$.

As for sequences ending or starting in an element $M \notin \mathcal{M}$, the only indecomposable there is S_2 . And the only sequence starting in S_2 (respectively ending in S_2) is $S_2 \xrightarrow{i_2} P_1 \xrightarrow{p_1} I_1$ (respectively $P_3 \xrightarrow{i_3} P_2 \xrightarrow{p_2} S_2$). But both of these sequences are almost split, and consequently not split. So they are non-zero as elements of $\text{Ext}_\Lambda^1(I_1, S_2)$ (respectively $\text{Ext}_\Lambda^1(S_2, P_3)$). The same argument holds for any non-indecomposable element as well, by simply reducing to sequences over indecomposables.

Having successfully proven that, this gives us the first example of a 2-exact category, with the simplest 2-exact sequence given by

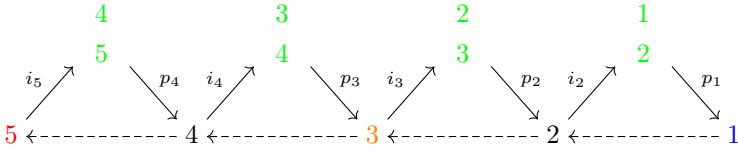
$$P_3 \xrightarrow{i_3} P_2 \xrightarrow{i_2 p_2} P_1 \xrightarrow{p_1} I_1.$$

To see this is 2-exact, one must check that $(i_3, i_2 p_2)$ is a 2-kernel of p_1 , and that $(i_2 p_2, p_1)$ is a 2-cokernel of i_3 . We will check the first statement, the second one being dual. Consider a map $f : C \rightarrow P_1$ such that $p_1 f = 0$ and $C \in \text{mod } \Lambda$ is an indecomposable module (SHOULD THIS BE ONLY FOR $\mathcal{M}$??). Then its image is either S_2 , or 0. If its image is 0, one can simply take $\hat{f} : C \rightarrow P_2$ to be the zero-map. If its image is S_2 , then C is either S_2 or P_2 . In the first case, that means that is (equivalent to) id_{S_2} , which is an issue. If $C = P_2$, then $f = i_2 p_2$ and one can simply take $\hat{f} = id_{P_2}$.

Next, consider a map $f : C \rightarrow P_2$ from an indecomposable module $C \in \text{mod } \Lambda$ such that $i_2 p_2 f = 0$. Since i_2 is mono, $p_2 f = 0$. But then since i_3 is a kernel of p_2 , there exists a unique map $\hat{f} : C \rightarrow P_3$ such that $i_3 \hat{f} = f$. This shows that $(i_3, i_2 p_2)$ is a 2-kernel of p_1 .

More generally, if one takes $\Lambda = k(1 \xrightarrow{\alpha_1} \dots \xrightarrow{\alpha_n} n+1) / \langle \{\alpha_{i+1} \alpha_i \mid i \in \{1, \dots, n\}\} \rangle$, we get a natural n -tilting subcategory, and therefore an example of an n -exact category. However, we may also get some lower exact structures as well.

Consider $\Lambda = k(1 \xrightarrow{\alpha_1} \dots \xrightarrow{\alpha_4} 5) / \langle \{\alpha_{i+1} \alpha_i \mid i \in \{1, \dots, 4\}\} \rangle$. The AR-quiver of this algebra is given by



The color code is the same as before, S_3 is colored orange since we will also consider it in our submodule category.

Let $\mathcal{M} \subseteq \text{mod } \Lambda$ be the full subcategory consisting of the additive closure of the colored indecomposable modules. Similarly to what was done before, we can prove that this is a 2-cluster tilting subcategory. This time, there are two natural 2-exact sequences: $X : P_5 \xrightarrow{i_5} P_4 \xrightarrow{i_4 p_4} P_3 \xrightarrow{p_3} S_3$ and $Y : S_3 \xrightarrow{i_3} P_2 \xrightarrow{i_2 p_2} P_1 \xrightarrow{p_1} S_1$. By 2.17, we also get for example that $X \oplus Y$ is 2-exact.

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